

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CASE STUDY (2 QUESTIONS)

Course Code: ITA05 **Course Title:** COMPUTER VISION

CO Assessed: CO3 – Precision (BL4) **Assessment:** Assessment Tool 1 – Case Study

Weightage: 40% of CIA **Total Marks:** 20 (2 Questions × 10 Marks)

CO1 Statement: Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4).

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Section / Batch: AI & ML

Date: 13-08-2026

Code of Conduct

I Nitheesh Kumar Kondapalli(192325090) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nitheesh Kumar Kondapalli

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus	Q Nos	Marks
Feature Detection and Matching	Preprocessing, Edge Detection, Harris Corner Detection, Hough Transform, SIFT	1	10
Feature Detection and Matching	PCA, Image Representation, Feature Matching, Image Processing Applications	2	10
GRAND TOTAL			20 Marks

ANSWER SCRIPT – ANNEXURE A CASE STUDY

CO3 – Precision (BL4) | Computer Vision (ITA05)

Question 1: Satellite Image Registration — Edge Detection vs Harris Corner Detection [10 Marks]

Answer structured per rubric: Understanding of Case (25%) → Application of Concepts (25%) → Analysis (20%) → Solution Design (20%) → Clarity (10%).

1. Understanding of the Case

The case describes a remote-sensing system that aligns (registers) satellite images of the same region captured at different times, in order to monitor environmental change. The original approach detects edges in each image and tries to match them across the time-separated pair. The system becomes unreliable because the two images differ in illumination (time of day / season / sun angle), orientation (satellite pass angle, slight rotation), and background content (seasonal vegetation, cloud cover, urban change). The core problem is that edges are a weak, low-information cue: an edge only marks where intensity changes, not a distinctive, locatable point, so edges from unrelated structures in the two images look alike and are easily mismatched under these varying conditions.

2. Application of Concepts

2.1 Why Edge Detection Fails Here

- Edges are 1-D, locally ambiguous cues — an edge pixel only encodes a strong gradient along one direction; long straight edges (roads, coastlines, field boundaries) look identical at many points along their length, causing the classic 'aperture problem' in matching.
- Sensitivity to illumination: shadows and sun-angle differences between the two capture times shift or create spurious edges (e.g., a shadow edge in one image with no counterpart in the other), directly corrupting correspondence.
- No rotation or scale handling: plain edge maps (e.g., from Sobel/Canny) carry no orientation or scale descriptor, so even a small satellite-pass rotation between the two images misaligns edge patterns.
- Background clutter: seasonal changes (crop growth, cloud shadows, water-level changes) generate new edges or remove old ones, so edge-to-edge matching has no way to distinguish a genuinely corresponding feature from an unrelated new edge.

2.2 How Harris Corner Detection Improves the Situation

Harris Corner Detection locates points where the local intensity gradient is strong in two independent directions simultaneously (i.e., true corners, not just edges), using the structure (second-moment) matrix M built from the image gradients I_x and I_y , and the corneriness score $R = \det(M) - k \cdot (\text{trace}(M))^2$. This resolves the case's problems as follows:

1. Distinctive localization: unlike a point on a long edge, a corner is uniquely localizable in two dimensions, so a corner detected on a field boundary intersection or building outline gives an unambiguous reference point for matching rather than an ambiguous point along a line.
2. Rotation invariance: the Harris response is based on the eigenvalues of M , which are invariant to image rotation, so corners remain detectable even when the second satellite pass introduces a small orientation difference.
3. Better robustness to illumination changes: because corneriness depends on the local gradient structure (relative change) rather than absolute intensity, moderate illumination and contrast differences between the two acquisition times affect corner locations far less than they affect raw edge maps.
4. Sparser, higher-quality candidate set: Harris returns a much smaller set of stable, high-confidence points compared to a dense edge map, which reduces the search space and the chance of false matches from clutter or seasonal background change.

5. Supports geometric verification: because corners are point features with (x, y) coordinates, a matched set of corner correspondences can be passed to a robust geometric model estimator (e.g., RANSAC-based affine/homography fitting) to compute the alignment transform and reject outlier matches — something not directly possible with raw edge pixels.

3. Analysis

The underlying reason Harris outperforms edge-based matching in this scenario is a difference in what is being matched: edges encode where intensity changes, while corners encode where intensity changes in a locally unique way. Image registration fundamentally needs unique, repeatable landmark points, not just boundaries, because the alignment transform (translation, rotation, scale) must be estimated from correspondences between specific locations, not general boundary shapes.

It is worth noting Harris alone is still not scale-invariant — if the two satellite passes differ significantly in altitude/zoom, Harris corners will not repeat reliably across scales, which is a known limitation. This is precisely the gap that motivates moving further to scale-invariant descriptors such as SIFT (as analyzed in the Assessment Tool 1 experiments) if the application also involves significant scale variation between captures.

Additionally, Harris corner matching still requires a descriptor or correlation window around each corner (e.g., normalized cross-correlation of a local patch) to establish correspondence between the two images; Harris itself only detects the interest points, it does not describe them for matching. This should be explicitly implemented alongside the detector.

4. Solution Design

1. Preprocess both images with radiometric normalization (histogram matching or CLAHE) so absolute illumination differences are reduced before feature detection.
2. Apply Harris Corner Detection independently on both time-separated images to obtain a sparse set of stable candidate points.
3. Describe each corner using a local patch descriptor (normalized cross-correlation window, or a rotation-aware descriptor) and match corners between the two images using similarity/distance thresholds.
4. Apply RANSAC to fit a global geometric transform (affine or homography) from the matched corner pairs, automatically discarding outlier matches caused by clutter or seasonal change.
5. Warp one image onto the other using the estimated transform to complete registration, and if scale differences are significant between passes, upgrade the detector to a scale-invariant method such as SIFT for the corner/keypoint stage.

This design directly targets the three failure modes named in the case — illumination, orientation, and background clutter — through radiometric normalization, rotation-invariant corner detection, and RANSAC-based outlier rejection respectively.

5. Clarity — Summary

Aspect	Edge Detection (original approach)	Harris Corner Detection (proposed)
Feature type	1-D boundary pixels	2-D distinctive corner points
Illumination robustness	Low — shadows create spurious edges	Moderate–High — based on relative gradient structure
Rotation handling	None	Invariant (via eigenvalue analysis of M)
Matching ambiguity	High (aperture problem along lines)	Low (locally unique points)

Aspect	Edge Detection (original approach)	Harris Corner Detection (proposed)
Supports outlier rejection (RANSAC)	Difficult	Well-suited (point correspondences)

Question 2: Image Representation for Real-Time Vehicle/Pedestrian Detection [10 Marks]

Answer structured per rubric: Understanding of Case (25%) → Application of Concepts (25%) → Analysis (20%) → Solution Design (20%) → Clarity (10%).

1. Understanding of the Case

An autonomous surveillance system captures high-resolution color video to detect vehicles and pedestrians in real time. Processing full color images directly (typically 3 channels — R, G, B — per pixel) increases the volume of data that every subsequent stage (filtering, edge detection, feature extraction) must process, which slows the pipeline below the required real-time frame rate. The engineering question is whether converting images to grayscale or binary before feature detection can restore the needed speed without unacceptably harming detection accuracy.

2. Application of Concepts

2.1 Color Representation

A color image stores three 8-bit channels per pixel (24 bits/pixel typically), so any per-pixel operation (convolution, gradient computation) costs roughly 3× the arithmetic of an equivalent single-channel operation, and memory bandwidth/cache pressure also roughly triples. Color does carry genuinely useful cues — e.g., separating a red brake light or a specific vehicle color from the background — but most classical feature detectors (Harris, SIFT, Canny) are defined on a single intensity channel and are typically applied to a converted grayscale version even when the source is color.

2.2 Grayscale Representation

Grayscale conversion (commonly $I = 0.299R + 0.587G + 0.114B$, weighting channels by human luminance perception) reduces the data to one channel per pixel. This cuts the computation for convolution-based operations (smoothing, Sobel/Canny gradients, Harris structure matrix) by roughly a factor of three, since gradient magnitude and orientation — the basis of virtually all classical edge and corner detectors — depend on intensity variation, which grayscale preserves well. This is why grayscale is the standard input for Harris, SIFT, and Canny in practice.

2.3 Binary Representation

Binary (thresholded, single-bit-per-pixel) representation goes further, reducing each pixel to foreground/background via a global or adaptive threshold (e.g., Otsu's method). This is extremely fast and memory-light, and is well suited to shape-based operations (contour extraction, connected-component analysis, morphological filtering), but it discards graded intensity information entirely. Fine texture, gradual shading, and weak-contrast edges — all important for distinguishing a pedestrian's silhouette from background clutter or separating two overlapping vehicles — are lost or merged, since binary thresholds are highly sensitive to lighting conditions and cannot represent partial/graded gradients that feature descriptors like SIFT rely on.

3. Analysis

The three representations sit on a clear efficiency-versus-information trade-off:

Representation	Relative Data / Compute	Edge Detection Quality	Feature Extraction Suitability
Color (RGB)	~3× baseline	Good, but redundant for gradient-based detectors	High information, high cost; needed only when color cues (e.g., signal lights) matter

Representation	Relative Data / Compute	Edge Detection Quality	Feature Extraction Suitability
Grayscale	~1× baseline	Excellent — preserves full gradient magnitude/direction	Standard input for Harris, SIFT, Canny; strong accuracy-speed balance
Binary	<< 1× baseline (1 bit/pixel)	Poor — only hard foreground/background boundary, sensitive to threshold/lighting	Useful only for coarse shape/silhouette tasks, not fine feature descriptors

Grayscale removes the two redundant color channels that most classical detectors never use directly, so the computational saving is largely 'free' in terms of detection quality — the gradient information that Harris, Canny, and SIFT actually operate on is essentially fully preserved. Binary conversion saves further computation but at a real cost: thresholding is a lossy, non-linear operation that erases the graded intensity transitions that gradient-based detectors and descriptors require, so detection accuracy for object categories with soft or variable-contrast boundaries (a pedestrian against a similarly-lit background, for instance) degrades noticeably, and binary thresholds are themselves unstable outdoors as lighting changes through the day — precisely the operating condition of a continuous surveillance system.

4. Solution Design — Recommended Representation

Grayscale is the most appropriate representation for this application, for the following reasons:

1. It removes the ~3× redundant channel cost of color while preserving the gradient magnitude and direction information that Harris corner detection, Canny edge detection, and SIFT/feature descriptors are built on, directly restoring the real-time frame rate the case requires.
2. Unlike binary conversion, grayscale keeps continuous, graded intensity values, so weak or low-contrast object boundaries (partially occluded pedestrians, vehicles in shadow) remain detectable rather than being merged into background by a hard threshold.
3. Grayscale is robust to a wider range of illumination conditions than a fixed/adaptive binary threshold, which is important for a system that operates continuously across changing outdoor lighting.
4. If specific color cues are operationally important (e.g., distinguishing brake-light activation or a specific alert-colored uniform), a hybrid design can be used: run the primary detection pipeline (edge/corner/feature extraction, tracking) on the grayscale channel for speed, while retaining the original color frame only for a lightweight secondary check (color-thresholding a small region-of-interest already located by the grayscale pipeline) rather than processing full-frame color at every stage.
5. Binary representation can still be used downstream, after detection, for fast morphological operations such as isolating a detected object's silhouette or computing its bounding contour once the region of interest is already small — but it should not replace grayscale as the input to the core feature-extraction stage.

5. Clarity — Summary

Converting from color to grayscale before feature detection gives the largest practical speed gain relative to information lost, because classical detectors are gradient-based and color's extra channels are largely redundant for that purpose. Binary representation offers even greater speed but sacrifices the graded intensity information needed for reliable, lighting-robust object boundaries, making it unsuitable as the primary representation for real-time, accuracy-sensitive vehicle and pedestrian detection. Grayscale, optionally paired with a targeted, small-scale use of color for specific cues, is therefore the recommended representation for this surveillance application.